

Free surface features of submarines moving underwater: Study of Bernoulli Hump

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ABSTRACT- Submarines are known to produce certain surface manifestations, primary among them being a feature known as Bernoulli Hump (BH). BH is a rise and fall of sea surface height above the submarine, when the vessel cruises close to surface. This feature has been noted by many researchers but scant information is publically available on the same due to the classified nature of the data. A major part of the Wide Area Surveillance program for submarines of the world navies uses the detection of this BH. In this work, our objective is to detect and establish the presence of BH characteristics. After calibrating a known CFD code using experimental data, we make model simulations which clearly depict the BH and its features. We then show the path ahead for submarine detection using BH from observations of sea surface heights. Value of BH height for a submarine at 10m depth moving at 5 knots is about 20cm which falls within the 2 cm vertical resolution of altimeters. Though most of the studies were for calm waters, an introduction to the presence of regular wind waves is given at the end.

KEYWORDS: *WAS, SAR, CFD, submarines and wake signature.*

NOMENCLATURE

WAS	Wide Area Surveillance
SAR	Synthetic Aperture Radar

CFD	Computational Fluid Dynamics
FFT	Fast Fourier Transform
RANS	Reynolds Averaged Navier Stokes
VOF	Volume of Fluid
SIMPLE	Semi-implicit method for pressure linked equation
ADV	Acoustic Doppler Velocimeter
L	Length of submarine
B	Beam of submarine
D	Depth of operation
k	Turbulence kinetic energy
ϵ	Dissipation rate of turbulence kinetic energy
V	Submarine forward velocity
G	Gravitational acceleration

1. INTRODUCTION

WAS is nowadays a terminology routinely used for national security. WAS indicates the surveillance of large swaths of area using Synthetic Aperture Radar (SAR) images from air-borne/space borne satellites (Kanjir et al., 2018). Western nations have been using this technology to detect unwarranted moving of targets in land and water. Most of this air-surveillance was done by Army to keep up with the transport and movement of trucks and other heavy enemy vehicles on land. Then later, this technology made its foray into the marine surveillance whereby SAR images are used to track ships/unauthorized vessels at sea (Deng et al., 2018).

Similar technique can also be used to track submarines. When a submarine moves at 40 m depth or less, it leaves a trail in the surface above. Very little work is done on this surface manifestation, the biggest contribution came from the work of Tunaley, 2015. He has used a Rankine ovoid method and by describing the submarine as a collection of sources and sinks, predicted the presence of Bernoulli Hump. But after that, literature is scant on this area, probably due to the sensitive nature of the data. In spite of the fact that the submarine itself can be detected directly, there are two main reasons why it is a better idea to detect the vessel's wake and not the vessel itself: the wake pattern is longer and clearer than the submarine itself in the image, and it can yield a better estimate of the submarine's actual position and velocity (Chen et al., 2008).

From the historical perspective, the trail left by ducks/boats when they sail through water has fascinated man because of the beautiful wake signature behind it (Thomson W, 1887). It was Lord Kelvin, who put scientific rigor to the phenomenon by demonstrating that the angle made by the wake of any object moving in water is the same, irrespective of the object itself. This wake now called as Kelvin wake is a combination of two wave fields- transverse waves and divergent waves. This wave system is found to converge at an angle of 39° (Raboud et al., 2013) at the position of ship.

Synthetic Aperture Radar (SAR) images of ship wakes show a wake in the form of dark and bright streaks (Tunaley et al., 1991). Dark streaks represent the Kelvin wake and bright streaks represent turbulent wake (Fig 1).

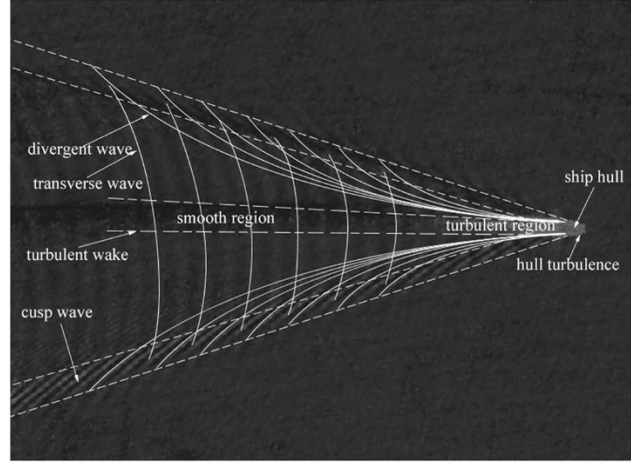


Fig. 1. Ship wake signature

Initially considered as a feature of beauty, it was later recognised that a wake signature is a source of hidden information, helping detection of ship's particulars (Peregrine et al 1971). Detection by extracting the information from Kelvin wake first was investigated by Newman, 1991. This was followed up by Griffins (Griffins et al., 1996). While Newman deals with estimates of the ship hull cross-section parameters (not the length), in Griffins only the estimates of the ship length are examined. An algebraic expansion for the width of the turbulent wake aft a cargo ship with zero axial net momentum was predicted in (Birkhoff et al, 1957), but only after quite some time it has been verified for other vessels (Peltzer et al., 1991). Unfortunately, the simple formula given in (Birkhoff et al, 1957) for the width of the turbulent wake aft a ship is purely asymptotic, i.e., it does not have a length-scale and, therefore, cannot be used for estimating the ship particulars. By invoking (Birkhoff et al, 1957) hypothesis, (Zilman et al., 1999, Zilman et al., 2004) derived a functional relation between the ship breadth and the width of its turbulent wake.

On the other hand, a lot of study on underwater vessels center around AUVs. Numerous authors have studied underwater bodies for understanding the hydrodynamic forces and moments on it (Jagadeesh et al., 2009; Saeidinezhad et al., 2015). Jagadeesh et al., 2009 has made experimental studies of hydrodynamic derivatives and other coefficients of an AUV hull form. Saeidinezhad et al. (2015) has studied the pitching of AUVs and drawn out the hydrodynamic coefficients associated with it. A lot of recent studies have been using CFD high fidelity simulations to capture the flow around the hull. The accuracy of turbulence model, the discretization of the equations and meshing thus play a vital role here (Sakthivel et al. 2011).

Most of the studies on submarines have been done for maneuvering and measurements of hydrodynamic derivatives. This is mostly because of the importance of proper maneuvering characteristics for submarines (Feldman, 1987, Quick et al., 2014, Dubbioso et al 2016). People have done captive model tests, PMM tests and rotating arm tests to measure these coefficients. Recently, there has been a surge in the application of CFD simulations in these tests. Tyagi and Sen, 2006 employed a RANS solver to study the horizontal hydrodynamic coefficients of an AUV. Wu et al., 2005 used a SUBOFF model to simulate a steady line motion with and without an angle of attack. The near-bottom effects on the hydrodynamic characteristics were investigated. Hu et al. 2007 used the CFX software to simulate the maneuvering tests by using the k- ϵ model to simulate the hydrodynamic derivatives.

Broglia et al. 2007, used CFD simulations to study the sway characteristics of a KVLCC tanker with free surface effects. The motion of the vessel was simulated using an overlapping mesh method with 8 blocks for a fixed background and 20 fitted blocks moving with the hull. The computed lateral force and the yaw moment agreed well with the experimental data with a relative error less than 5.5% and 20%, respectively. Yang et al., 2011 investigated a similar phenomenon without considering free surface effects.

This paper aims to study a possible detection technique using CFD software. It may be possible to detect a submarine using accurate measurements of the ocean surface height or by imaging the water surface. The major physical feature will be the wake developed by the underwater body in motion. Three different phenomena are caused either directly or indirectly by surface wakes: internal waves and turbulence, Kelvin wake and Bernoulli's Hump. In this study Bernoulli's hump is the main area of focus. Some of the original works made by (Hershey, 1966) is based on Rankine ovoid model, which is also called "Doublet" based on single source and sink. The same methodology is applied by (Stefanick, 1987) in which he prepared a table for various size of Bernoulli's hump created by various submarine types. Later (Daly, 1994) produced this table in his study of submarine detectability.

Computational Fluid Dynamics (CFD) is a vast area of research that was traditionally in the Mechanical Engineering discipline. But in the last few decades, there has been a massive breakthrough in practical applications of CFD towards ship hydrodynamics (Zhang et al., 2010; Kundu et al., 2019). CFD can play a valuable role in the investigation and prediction of ship performance. Model testing, which has always been the traditional technique for accurate

prediction of any ship flow, is more costly and time-consuming as compared to computational analysis. In the present day, many researchers are using CFD as a tool for better predictions of almost all ship motions including maneuverability (Kundu et al., 2019) and propulsion. In Barman et al. 2018, a RANS code was used to make predictions of ship wakes and a large database of high fidelity wake data was prepared for reference. In that paper, a numerical investigation into detecting the ship velocity was also carried out. Ship's velocity prediction is based on the fact that the transverse wave produced by ship moves at the velocity of the ship itself (Tunaley et al., 2003). In the current work, we use CFD for investigating the movement of underwater vessels and study the sensitivity of Bernoulli Hump to depth of operation and submarine velocity. Then we investigate this predictability in the presence of wind waves.

For validation and verification of the numerical simulation results, experiments were conducted to get submarine wake signature, as in Cimbala et al, 2013. Experimental setup was arranged in our departmental hydrodynamic laboratory. A particular vessel of a cylinder-like shape was used to calibrate a STARCCM+ model (a commercial code owned by Siemens version 2019.3) . Then numerical analysis was performed using that calibrated model to simulate various test cases of depth of operation and velocity of submarine models.

2. MATERIALS AND METHODS

2.1 Background

The prescribed method of detecting underwater bodies is by two techniques: direct and indirect techniques. In direct technique, the vessel itself is detected; whereas in indirect method, any physical or environmental disturbance caused by the vessel is detected as a proxy for the vessel itself.

Direct Detection: In this case, the light reflected from the submerged vessel is used to detect the vessel. Usually the blue-green (450-550nm) band of the spectrum is used, as it is less absorbed by ocean itself.

Indirect Detection: All other methods of detection are by indirect means. These are classified into: physical surface effects, optical effects and thermal effects.

Physical Surface Effects: The major physical surface effect is to detect the wake due to the vessel. The wake can be classified into Kelvin wake, Bernoulli Hump and internal waves.

The Bernoulli Hump: When the submarine travels underwater, it produces a vivid rise and fall of sea surface height above it. This feature is known as Bernoulli Hump and it is a function of velocity, depth of operation and dimensions of the vessel. The height of the hump can vary from tens of centimeters near the surface to a few mm at a depth of about 50 m (Stephanick et al. 1988).

Kelvin Waves: These waves were discussed in detail in the introduction section and we donot elaborate it here again.

2.2 Equations

The RANS equations are classical time-averaged forms of Navier-Stokes equations (Wilcox, 1993)The Navier-Stokes equation for fluid motion is given as,

$$\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho} \frac{\partial p}{\partial x_i} + F_i + \nu \frac{\partial^2 u_i}{\partial x_j \partial x_j} \quad (1).$$

moreover, the incompressible continuity equation is

$$\frac{\partial u_i}{\partial x_i} = 0 \quad (2)$$

Where $u(x, t)$ is the velocity component in i^{th} the direction at a definite point in space x and time t , p is the static pressure, ρ is the density of fluid, ν is the kinematic viscosity and F_i is an external force in the i^{th} direction. To see how turbulence coexists with mean flow, it is convenient to analyze the flow in two parts, a mean (or average) component and a fluctuating component. The equations are therefore rewritten to remove all fluctuations arising from turbulence. Thus, the instantaneous velocity, pressure and stresses can be written as:

$$u_i = \overline{u_i} + u_i' \quad \sigma_{ij} = \overline{\sigma_{ij}} + \sigma_{ij}' \quad (3)$$

Introducing the above to the earlier equations and averaging, we obtain the RANS equations (for incompressible fluid) as,

$$\frac{\partial \overline{u_i}}{\partial t} + \frac{\partial}{\partial x_j} (\overline{u_i u_j}) + \frac{\partial}{\partial x_j} (\overline{u_i' u_j'}) = -\frac{1}{\rho} \frac{\partial P}{\partial x_i} + F_i + \frac{1}{\rho} \frac{\partial \overline{\sigma_{ij}}}{\partial x_j} \quad (4)$$

Or,

$$\frac{\partial \overline{u_i}}{\partial t} + \frac{\partial}{\partial x_j} (\overline{u_i u_j}) = -\frac{1}{\rho} \frac{\partial P}{\partial x_i} + F_i + \frac{1}{\rho} \frac{\partial}{\partial x_j} (\overline{\sigma_{ij}} + R_{ij}) \quad (5)$$

$$\text{Where } R_{ij} = -\rho \overline{u_i' u_j'}$$

This is interpreted as additional stress known as Reynolds stress. The RANS equations in three spatial dimensions and a mass continuity equation give four independent equations with ten unknown variables. The presence of Reynolds stresses $(\overline{\rho u_i' u_j'})$ makes closing the turbulence problem difficult. Calculation of R_{ij} requires a turbulence modeling and in our simulations we adopt k- ϵ model. Equations 1-5 are referenced from Wilcox, 1993.

Numerical simulations were performed for these equations using STARCCM+ to study the wake formation on the surface of the water when submarine moves underneath. These RANS simulations employ Volume of Fluid method for multiphase flows, to capture the disturbance at interface of air and water, accurately. The STARCCM+ software is used with the $k-\epsilon$ turbulence model to solve the transport equations for turbulent kinetic energy and the rate of dissipation of turbulent kinetic energy. Semi-implicit method for pressure linked equation scheme(SIMPLE) is used to solve the discretized equations.

Numerical simulations were performed on bare hull, for investigation of submarine signatures at a velocity of 5 and 10 knots (2.5 and 5 m/s). The time step size for convergence of the solution has been taken 0.01 sec. Once numerical solutions converged, disturbance caused by moving vessel was captured in the imagery. These images helped in developing the complete picture of the Bernoulli Hump.

2.3 Models used for testing:

Though for calibration, we used a particular model (shown in Fig 4), for testing two different models were used for the major part of the study. The profiles of both the tested models are similar, with a cylindrical body having a hemispherical front end and a conical tail. The first model is a full-scale model absorbing the basic dimensions of length and beam from the Akula class submarines (113.3 m long and 13.6 m beam) of the Indian Navy (Fig 3), while the second model has the dimensions from the Sindhughosh class submarines (72.6m long and 9.9m beam).

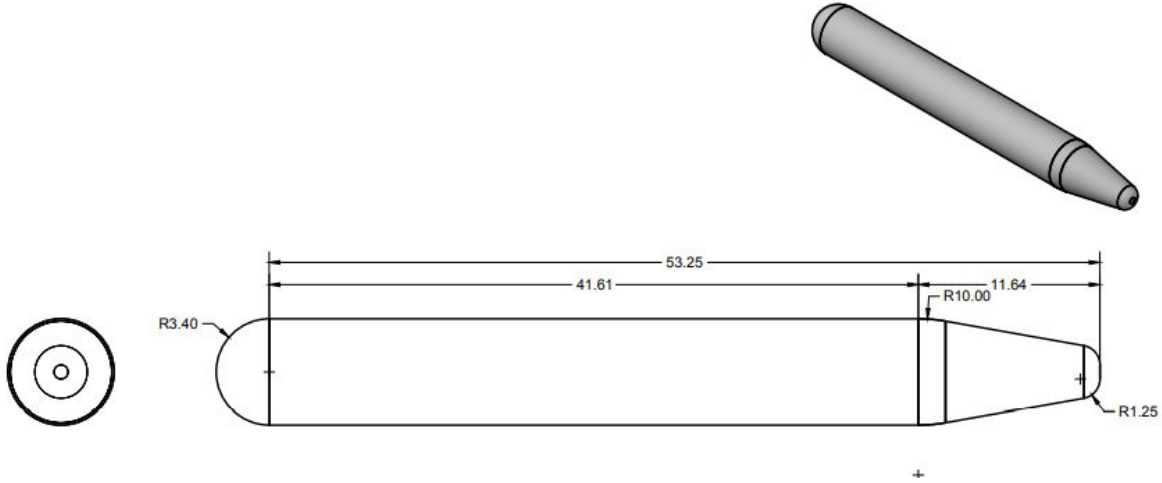


Fig 2: Model with dimensions of Akula class submarine drawn at 1:2 scale.

Meshing is done in such a way that cells are carefully distributed in the region surrounding the submarine. The complete analysis is done in full-scale. The base size of the grid is set at $L/50$, where L is the length of the submarine. The mesh size has been fixed after a series of mesh convergence iterations performed. This mesh size is as recommended by the ITTC standards (see Fig 2b). Trimmed cell mesher is a volume mesher and is primarily used to discretize the domain. Prism Layers are used around the submarine body in order to capture the boundary layer that is formed submarine, by the virtue of viscosity. The free surface is also discretized into smaller cells to capture the free surface elevation distinctly. For different depths of operations, the refinement regions and submarine model are translated downwards. Refined meshes are provided along the pressure points of the submarine to capture the flow variations induced by the submarine. Sufficient damping factors are provided from the boundaries until the refined mesh spaces so as to dampen any unnecessary wave reflections. Thus, we have a total of about 2 to 2.5 million cells, varying based on the different models employed and different operating conditions.

Number of meshing elements and number of nodes were 2.6×10^6 and 5.0×10^5 respectively. The values of y^+ for these simulations are given in Table 2a. Under mesh quality, we have tried to keep element quality, aspect ratio, orthogonality and skewness within acceptable range for better predictions. Table 1 details the mesh quality.

Table 1. Meshing parameters

Mesh quality	submarine
Orthogonal	0.76
Skewness	0.22
Aspect ratio	1.87
Element quality	0.82

Testing of grid for numerical accuracy was done through grid-independence test. This test has been done for three different cases of mesh quality (coarse, medium and fine). Meshing elements and nodes are mentioned in Table 2a.

Numerical simulations were performed for the three different mesh sizes for checking grid independence. The grid convergence study was performed for detected (backward calculation from surface pressure) velocities of the submarine moving underneath. The numerical velocity of the model was set from 0.5 to 5 m/s with the submarine surfaced. From the pressure profile (using elevations) the velocity can be detected through a method given in Barman et al., 2018 (from freesurface pressure distribution). So indirectly, we can say that we are using the values of freesurface pressure distribution to do the grid convergence. Numerical solution for fine mesh gives the most accurate detected trend of forward velocity with less than 10 percentage error (Table 2a).

Table 2a. Grid independence test for coarse, medium and fine mesh at $v=5\text{m/s}$

Parameters	Coarse (h_1)	Medium (h_2)	Fine (h_3)
Number of elements	710927	946273	1286928
Number of nodes	150499	183045	248661
y^+	98.4	60.2	38.6
Detected forward velocity	4.552 m/s ($=\phi_1$)	4.570 m/s ($=\phi_2$)	4.590 m/s ($=\phi_3$)

For grid independence analysis, we see that the derived velocities of the submarine varies only varies less than 1% between the coarse and fine meshes (Table 2a.). Thus the coarse resolution is sufficient for numerical accuracy.

The grid dependency analysis was also performed as per the method suggested by Celik et al. 2008. The method is mainly based on Richardson extrapolation and is a well-established practice. The purpose of this check is to ensure that the result obtained from the chosen grid lies in the asymptotic range of convergence. Typically, the mesh resolution used for the study is slightly coarse. In these simulations, only the velocity (given and derived when the submarine is surfaced) is used to calculate the grid convergence.

For the analysis, the grid refinement is performed in all x, y and z axis, maintaining same refinement ratio, about $\sqrt{2}$. The order of accuracy, p and Grid Convergence Index (GCI) are predicted using the following equations,

$$p = \frac{1}{\ln r_{21}} \left(\ln \left(\frac{\varepsilon_{32}}{\varepsilon_{31}} \right) + \ln \frac{(r_{21}^p - s)}{(r_{32}^p - s)} \right) \quad (6)$$

$$GCI_{fine}^{21} = 1.25 e_a^{21} / (r_{21}^p - 1) \quad (7)$$

In the equations, $s = \pm 1$, depending on the sign of the convergence. Parameters are given in Table 2b.

$R = GCI^{23} / GCI^{12} r^p (\approx 1 \text{ in the asymptotic region})$ (Eq. 6,7 referenced from ASME V&V 20, 2009)

As was seen from the analysis, the results showed both monotonous and oscillatory convergence. This is contributed to the very low mesh resolution used for the coarse mesh cases. Moreover, the target for the simulations was to properly capture the static pressure/elevations and velocities. The uncertainty (GCI) is low for velocity predictions. Overall, the applied coarse resolution was sufficient to attain reliable results with low uncertainty.

Table 2b: Uncertainty analysis for grids

Item		Velocity (m/s)
		5
Refinement ratio	$r_{21} = h_2/h_1$	1.33
	$r_{32} = h_3/h_2$	1.35

Convergence	$\varepsilon_{21}/\varepsilon_{32}$	0.9
Order of accuracy	P	0.28
Grid convergence index (GCI)	GCI_{fine}^{21}	0.017
	GCI_{fine}^{32}	0.015

All numerical simulations were done in order of 10^6 Reynolds's number. Different turbulence models were tested in simulations to better capture submarine wake signature. Tests were done with the k- ε turbulence model (Realizable and Standard condition) and k- ω turbulence model (SST and standard SST) for our present study. Since the results were identical for various models, the k- ε turbulence model was selected for numerical simulations.

The STARCCM+ numerical calculations were conducted in a single node Intel(R) Core i5 CPU with 7 cores, clock speed 2.27 GHz and 16 GB of physical memory. The average time step used was 0.01 s and the model was run till steady state conditions which was about 35 seconds. This took about 24 hours of clock time.

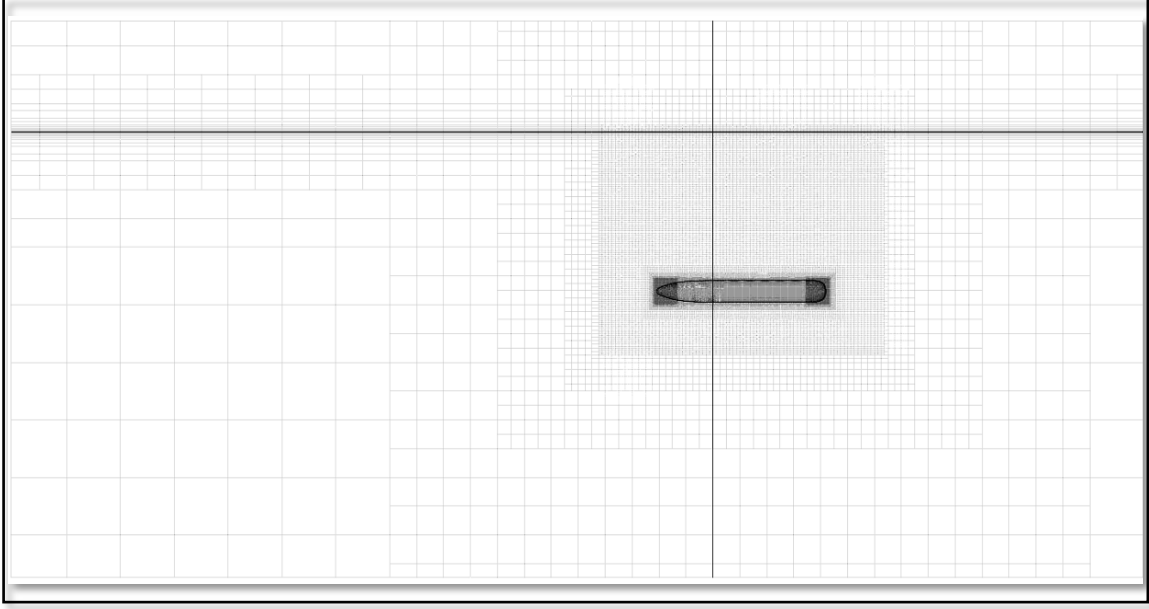
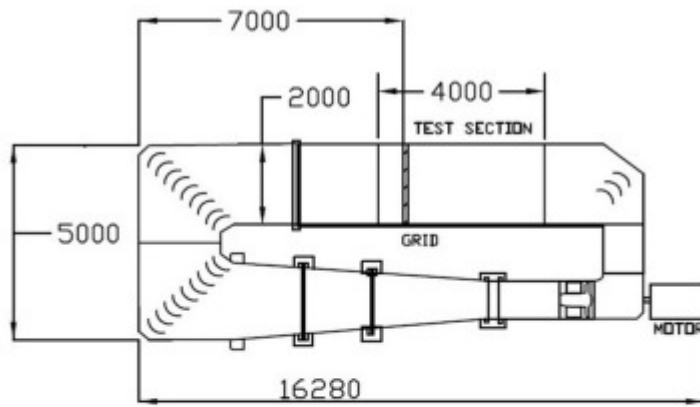


Fig 2b: Mesh for the submarine simulations

2.4 Experimental setup for calibration:

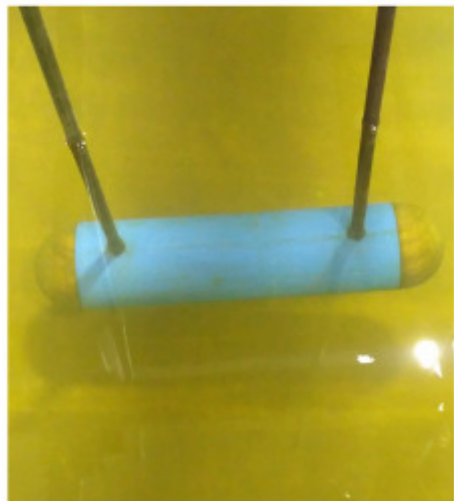
All experiments mentioned here were conducted in the recirculating water tank of IIT Kharagpur. The construction is of concrete with the side walls are made of glass for viewing the experiment (Mitra et al. 2019). The schematic of the setup with the position of grids and underwater vessel is shown in Fig 3a. The water is pumped by an electronic motor. We kept the model at a depth of 0.8 m where a mean flow velocity up to 1m/s is achievable. The water tank has width 2 m and depth 1.5 m. The experimental setup is shown in 3b.



(a) Schematic of the recirculating tank



(b) Grid setup in the tank



(c) Submarine model

Fig. 3. Experimental setup for the current work. The schematic diagram is reproduced from Panda et al., 2018.

Fig 3b shows the grid setup for generating turbulence by using PVC pipes. Size of the mesh is $M=0.1\text{m}$ and the diameter of the pipes is 0.025 m . Details can be found in Panda et al., 2018.

2.4.1 Model used for experiments and STARCCM+ calibration

The model is placed at a distance of 0.4m from the grid in the recirculating tank. A schematic of the setup for the experiment of the physical flow problem is shown in Fig 4. Direction of flow in Fig 4 is given by arrows. The underwater vessel hull is of cylindrical shape with hemispherical ends as shown in Fig 4. The length and diameter of the vessel are 0.5 and 0.1 m respectively. The radius of the hemispheres is equal to the radius of the cylinder. Hull shape is as described in Mansoorzadeh et al., 2014. Since the main aim was to study turbulent flow around an underwater vessel, we just assume the hull of the submarine and not control surfaces. More detailed study should include the propeller effects, sail, rudder surfaces and hydroplanes.

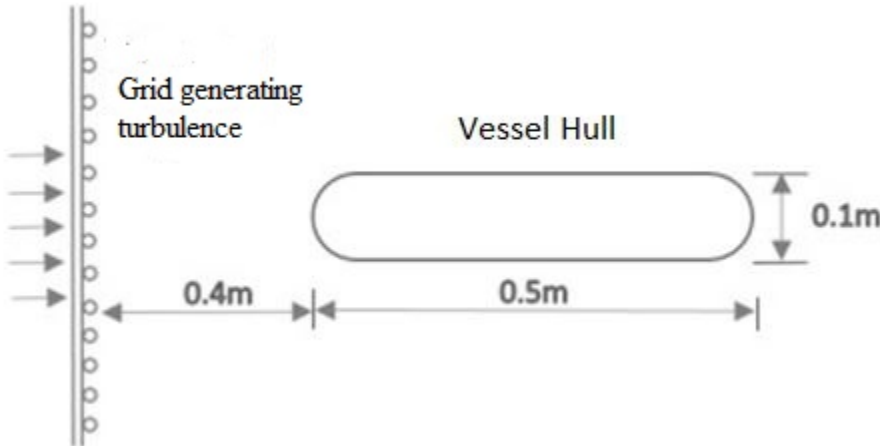


Fig. 4. Schematic of the problem setup. Arrows indicate direction of flow.

x, y, z represent the streamwise, transverse and vertical directions respectively. U, V and W are the longitudinal, transverse and vertical mean velocities in x, y and z directions respectively.

An ADV (Acoustic Doppler Velocimeter) is used for calculating the instantaneous velocities of the flow in all directions. The sampling frequency was set at 200 Hz . The ADV was mounted on a pulley and stayed at one point for measurements for about 5 minutes (Voulgariset al., 1998). A spatial and temporal resolution of 1cm^3 and 200Hz respectively is achievable through the ADV in these experiments.

An ADV measures the velocities based on the principle of Doppler shift by sampling at high frequency rates. The main elements of the ADV are a signal processing electronic module, acoustic receivers and acoustic transmitter. The schematic of the ADV is shown in Fig 5.

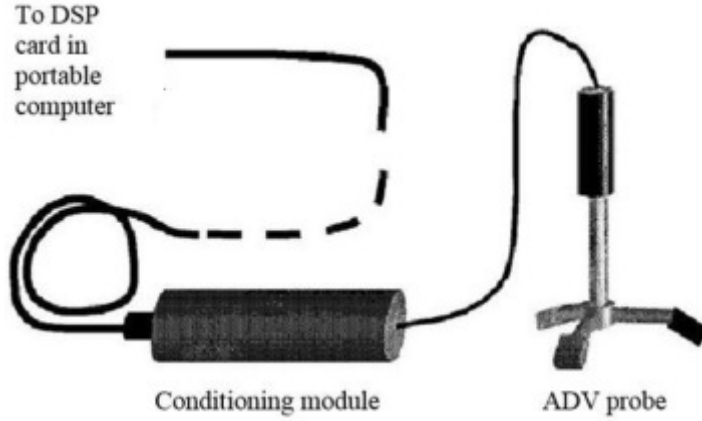


Fig. 5. Schematic diagram of ADV and conditioning module, reproduced from Panda et al.2018.

The errors associated with the measurement and sampling have been quantified (Benedict et al., 1996). Primary sources of error in the system are due to the placement of the probe, presence of fluid properties like dust and dirt in water and error in sampling frequencies. The mean velocities in ADV are sampled with less than 1% error (Voulgaris et al., 1998). We ran the ADV at the sampling frequency of 200 Hz where an accuracy of 0.5% with a value ± 1 mm/s were achieved.

The ADV measures the velocity data at different points downstream of the grid. These data are then decomposed into mean and fluctuating velocities as

$$u_i = \bar{u}_i + u'_i \quad (8)$$

$\bar{u}_i$ and u'_i can be calculated using the following formula:

$$\bar{u}_i = 1/n \sum_{i=1}^n u_i \quad (9)$$

$$u_i' = \sqrt{1/n \sum_{i=1}^n (u_i - \bar{u})^2} \quad (10)$$

(Equations 8-10 are referenced from Wilcox 1993).

2.4.2 Experimental uncertainty analysis

To predict the uncertainty of any experimental variable related to its component uncertainty, experimental uncertainty analyses need be performed. It is also known as a propagation of uncertainty (Cimbala et al., 2013).

In the present work of experimental uncertainty analysis, we have experimental variable Reynolds stress (RS) which is the function of velocity 'u' and 'v'. The main objective of the experimental uncertainty analysis is to calculate the uncertainty of 'RS' caused due to the uncertainty of u' and v'. Once the uncertainty analysis has been done, RS can be written as a RS = RS ± ΔRS where 'ΔRS' will be the experimental uncertainty of RS. For the investigation of experimental uncertainty, we used Root of the Sum of the Square uncertainty(RSS) uncertainty. It has a 95% uncertainty level for each of the uncertainties.

$$RS = \overline{u'v'} \sim u_{rms} \cdot v_{rms}$$

$$dRS = \Delta u_{rms} \frac{\partial RS}{\partial u_{rms}} + \Delta v_{rms} \frac{\partial RS}{\partial v_{rms}} = \sigma_{u_{rms}} \frac{\partial RS}{\partial u_{rms}} + \sigma_{v_{rms}} \frac{\partial RS}{\partial v_{rms}} \quad (7)$$

(Cimbala 2013), where dRS, $\sigma_{u_{rms}}$, $\sigma_{v_{rms}}$ are uncertainties in Reynolds stress, u' (taken as u_{rms}) and v' (taken as v_{rms}) respectively.

We did the experiments at three different velocity magnitudes of 0.6, 0.8, and 1.0 m/s (Table 2c). Relative uncertainty, also known as fractional uncertainty and its magnitude showed minimal effects on Reynolds stress (see table 2c). 0.33% is the maximum RSS uncertainty in RS which is negligible.

Table 2c. Experimental uncertainty analysis for three flow velocities (m/s)

velocity 'U' (m/s)	0.6	0.8	1.0
σ_{urms} (measured)	0.001	0.001	0.001
σ_{vrms} (measured)	0.001	0.001	0.001
Uncertainty in RS	0.0012	0.0016	0.002
Expected Uncertainty	0.36	0.64	1.0
'RS _{ex} '	± 0.0012	± 0.0016	± 0.002
Relative RSS uncertainty ($\Delta \text{RS}_{\text{rel}} = 1 - \text{RS} / \text{RS}_{\text{ex}}$) (%)	0.33	0.025	0.2

3. RESULTS

3.1 STARCCM+ Calibration

The primary problem of detecting the very faint Bernoulli Hump in the laboratory forced us to adopt the following method. We prepared the model of an underwater vessel as described in last section and tested it. We measured the variation of turbulent kinetic energy and normal and tangential Reynolds stresses. Then a STARCCM+ model was calibrated to those results (shown below). After this, we scaled up the numerical model to full-scale submarine dimensions and test-simulated it for various conditions. The Bernoulli Hump was noticed in the simulations above the position of the submarine. The model was tested for various depths of operation and velocities of operation. We present these results. Fig 6a-d shows the comparison of turbulent kinetic energy between model results and experimental values. The circles represent the experiment values whereas the curves represent STARCCM+ results (with and without wall reflection terms). As can be seen, the values are matching to high degree of accuracy. The x axis represents the non-dimensionalized (with radius of vessel) distance along the hull (readings just near the hull).

Model was calibrated for this experiment by using the STARCCM+ software. The numerical model was calibrated with a linear pressure strain model, for slow pressure strain correlations. Two cases were investigated, the presence of slow and rapid pressure strain and one more with the presence of wall reflection term. Fig 6a-d show the values of non-dimensionalized TKE and normal and tangential Reynolds stresses from both numerical and experimental methods. It was clear that including the wall reflection term improved the results (Gibson and Launder, 1978). Fig 6b shows how the energy transfer occurs between the stream-wise and transverse components of flow. So for all our submarine simulations also, we considered the wall reflection term near the surface.

The model was thus calibrated for domain of simulation, grids (meshes), turbulence model and boundary conditions, and then we chose to run for the two full-scale submarines.

3.1.1 Boundary conditions for CFD

While simulating in full scale, more number of cells are required to capture the Bernoulli hump. Since the submarine is symmetric about centreline, it is reasonable to assume that flow is symmetric. Hence only one half of the fluid domain is modeled. This reduces the number of cells resulting in reduced computational time. The domain and boundary conditions of full-scale simulation are shown in figure5.1.

The volume mesh consists of hexahedrons in an unstructured grid. The mesh is generated using the trimmer-mesh function in STAR-CCM+. Prism layer mesh is used on the hull surface in order to make a reasonable estimation of the shear stresses on the hull. Free-Surface is also refined to capture the Bernoulli hump.

Table 2d: Boundary conditions

Boundary	Boundary Condition	Domain Size
Inlet	Velocity Inlet	$3.0 \times L$ distance from the origin
Outlet	Pressure Outlet	$4.0 \times L$ distance from the origin
Side Wall	Velocity Inlet	$2.0 \times L$ distance from the origin
Top	Velocity Inlet	$0.6 \times L$ distance from the origin
Bottom	Velocity Inlet	$2.5 \times L$ distance from the origin
Symmetric side wall	Symmetry	Not Applicable

- L - Length of the model
- Origin is along the free surface

The boundary conditions provided to the different boundaries of the domain are shown in figure 5.1.

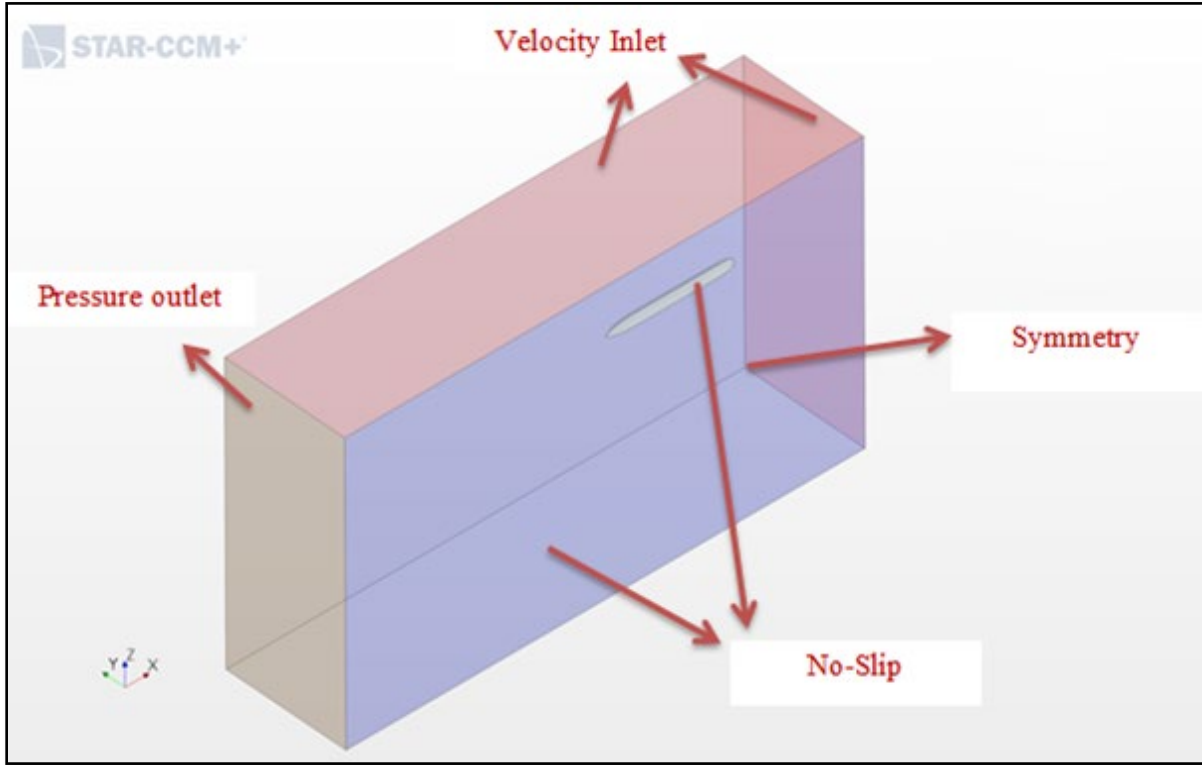


Figure 5.1: Domain considered in full-scale

3.1.2 Time step

A steady-state solver is used for solving the simulation model. A first-order discretization is employed. The time step is determined based on the ITTC guidelines as given below:

$$\Delta t < \frac{1}{200} \left(\frac{L}{U} \right)$$

The guideline mentioned above is applied and the time step of 0.01s is used for 10 knots simulations and 0.02s is used for 5 knots simulations. The number of inner iterations has been set to 10.

The under-relaxation factors are also set accordingly. A value of 0.7 is set for velocity, 0.4 for pressure, 0.8 for the k-epsilon turbulence, 1.0 for k-epsilon turbulent viscosity and 0.8 for segregated VOF.

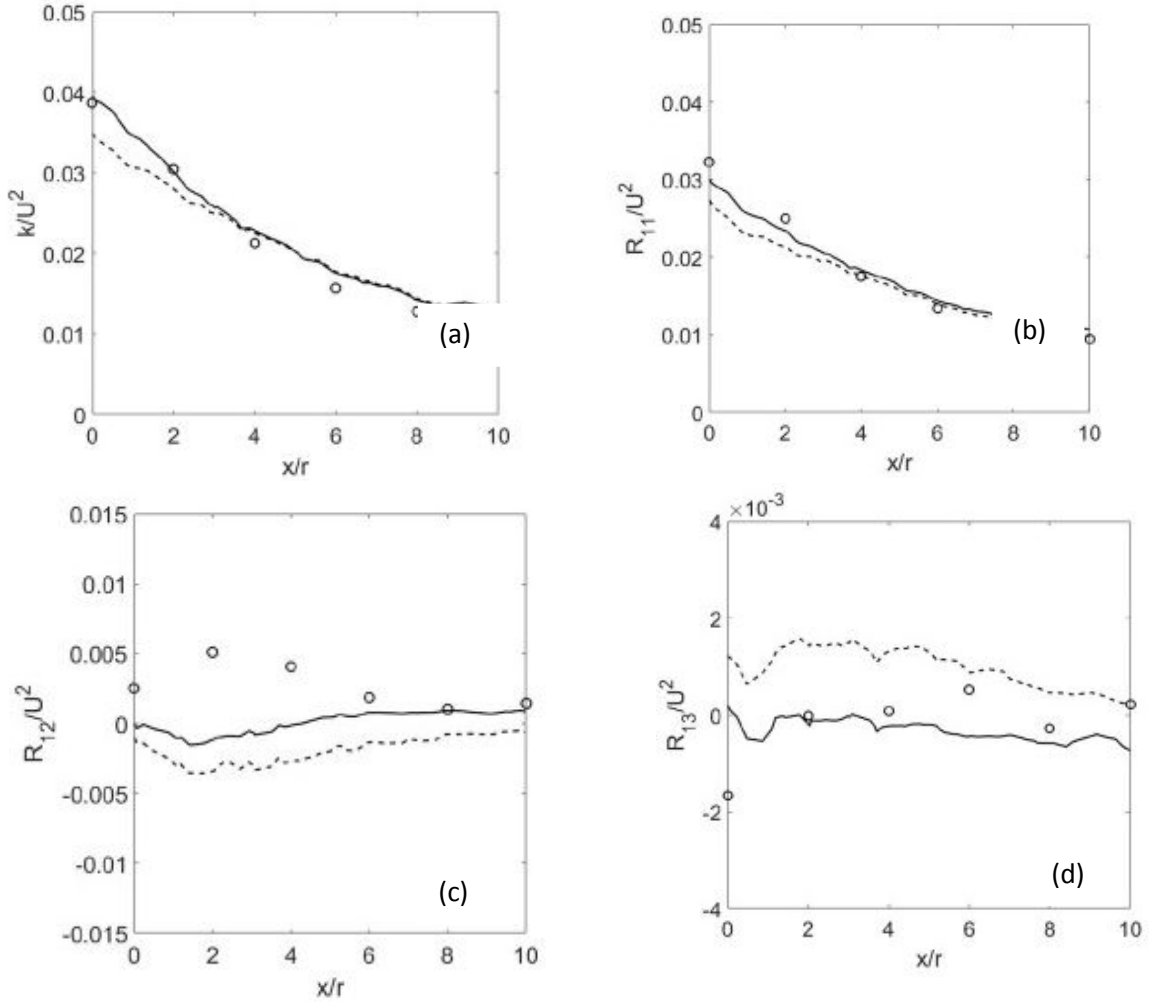


Fig 6a-d: Plot of the simulated and experimental values of kinetic energy and Reynolds stresses. The circles represent experimental values, the dashed lines (without wall reflection) and solid (with wall reflection) of STARCCM+ results.

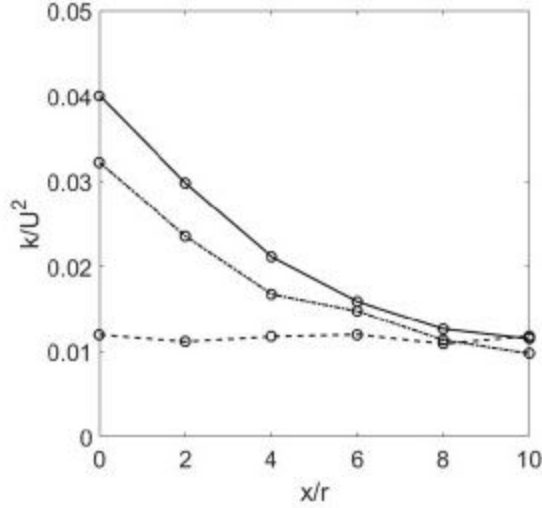


Fig 7: Experimental (circles) and STARCCM+ modeling results. Dashed line is for vessel without grid, dash-dotted curve is grid only and solid lines for (vessel + grid) case.

3.2 Testing:

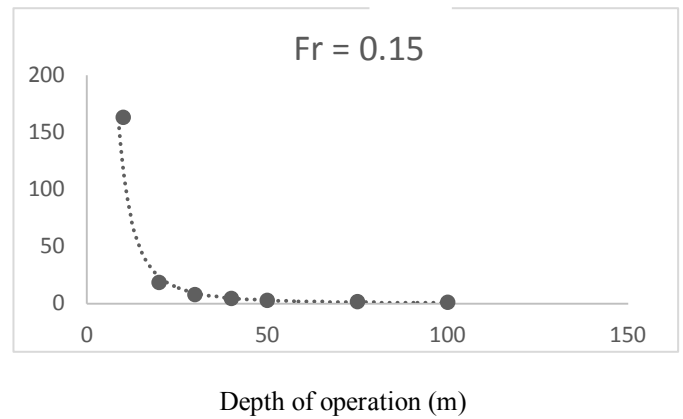
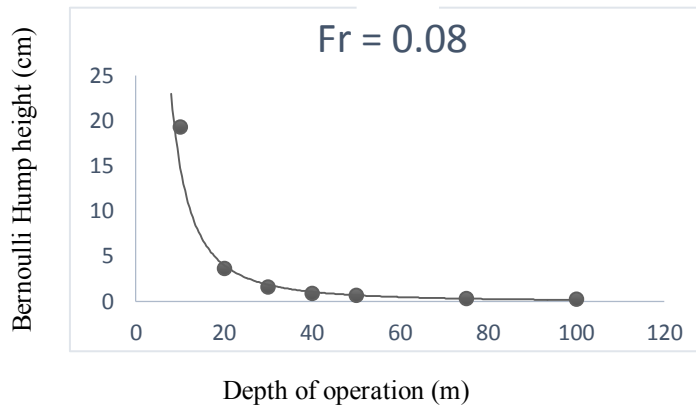
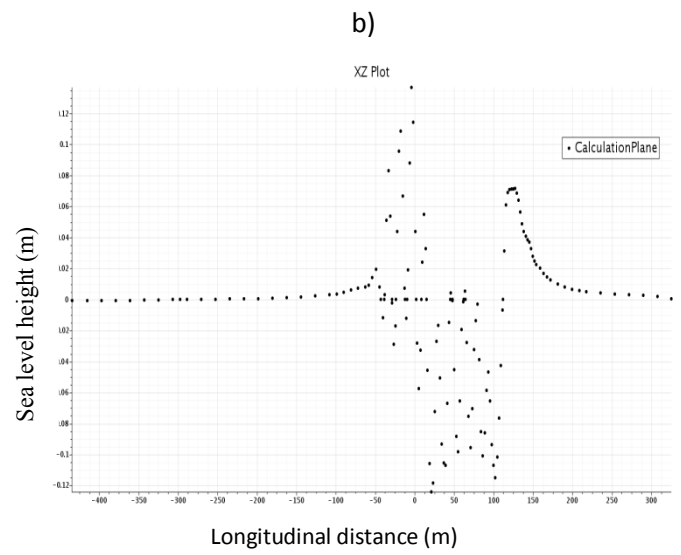
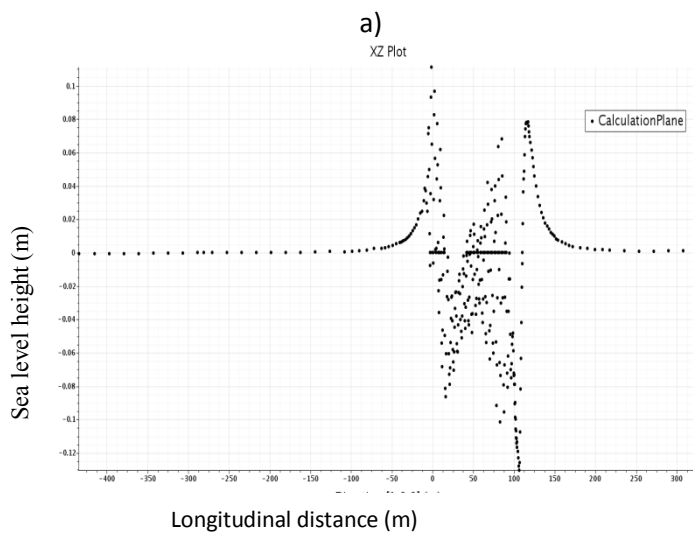
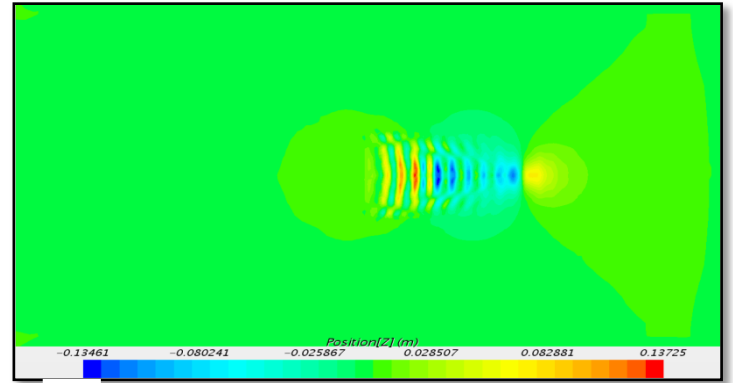
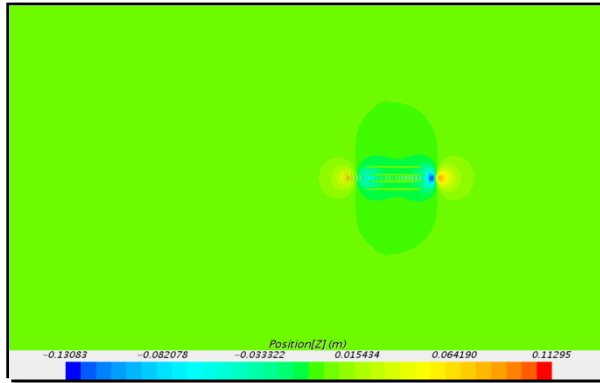
After calibrating the model domain, grids, turbulence model and boundary conditions, we have chosen to run for the model submarines. A full scale model was chosen for the simulations. An Akula submarine and Sindhughosh submarine were prepared as in Fig 2. These models were run for velocities of 5, 10 and 15 knots cruising speed. The models were placed 10,20,30,40,50 m below the free surface.

Table 3 shows the Akula submarine at a speed of 5 knots. As is expected, the BH height decreases as the depth of operation increases. At 10 m depth, the BH height is 19 cm while it reduces to 3 cm by 20 m depth of operation. It is a little dubious if the height of 3 cm could be detected. But with the high resolution altimeters with vertical resolutions of 2 cm or less, it might be possible. A very thorough filtering of the ocean surface is necessary to make this observation. But table 4 shows that at 10 knots speed, a submarine at a depth of even 40m might be detectable.

Table 5 shows similar results for the Sindhughosh submarines. At a speed of 5 knots, the submarine is visible at only 10 m depth. Beyond that BH goes into mm which is not detectable.

Table 3: Akula class submarines at 5 knots speed

Submarine Length		113.3 m	Froude Number 0.8
Submarine Beam		13.6 m	
Sl. No	Velocity Knots(m/s)	Depth (m)	Bernoulli Hump height (cm)
1	5 (2.57)	10	19.29
2	5 (2.57)	20	3.68
3	5 (2.57)	30	1.58
4	5 (2.57)	40	0.93
5	5 (2.57)	50	0.66
6	5 (2.57)	75	0.31
7	5 (2.57)	100	0.26



e)

f)

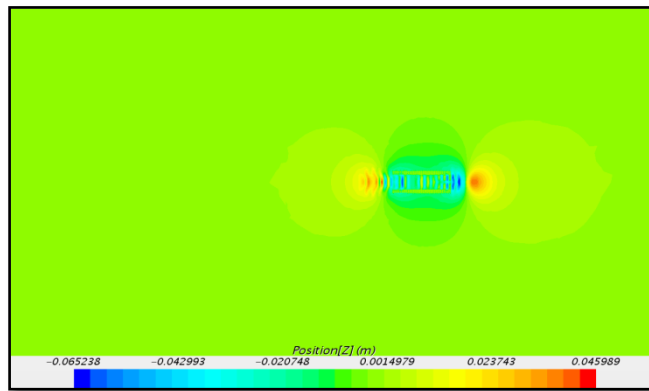
Fig 8a-f: BH (cm) for Akula class submarines at 5 knots (a,c,e) and 10knots (b,d,f) speed.

Table 4: Akula class submarines at 10 knots speed

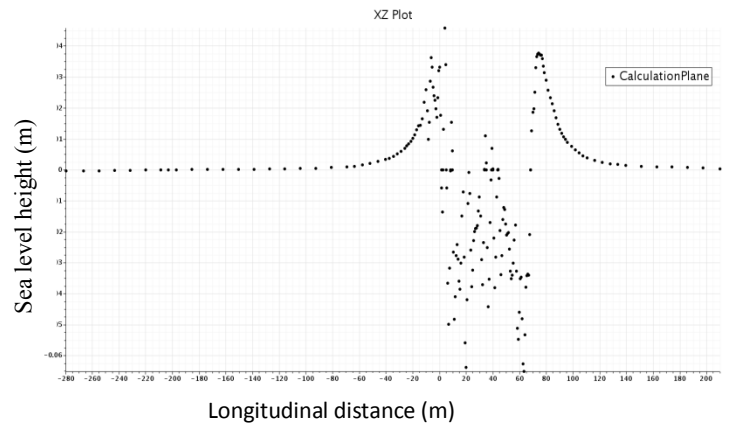
Submarine Length		113.3 m	Froude Number 0.15
Submarine Beam		13.6 m	
Sl. No	Velocity Knots(m/s)	Depth (m)	Bernoulli Hump height (cm)
1	10(5.14)	10	162.88
2	10(5.14)	20	18.23
3	10(5.14)	30	7.52
4	10(5.14)	40	4.02
5	10(5.14)	50	2.74
6	10(5.14)	75	1.25
7	10(5.14)	100	0.59

Table 5: Sindhughosh class submarines at 5 knots

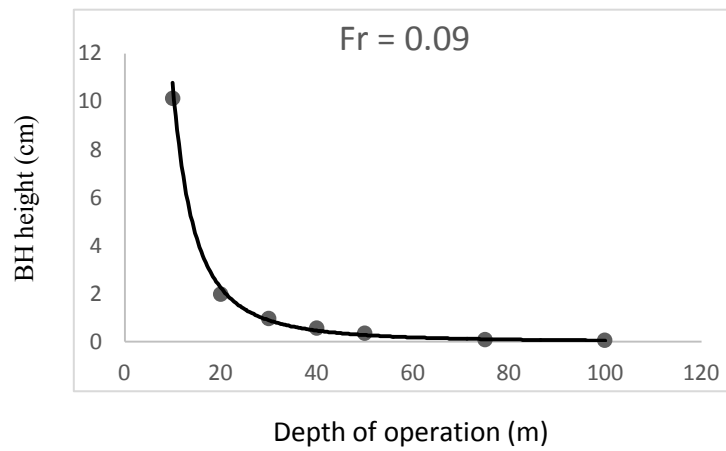
Submarine Length		72.6 m	Froude Number 0.09
Submarine Beam		9.9 m	
Sl. No	Velocity Knots(m/s)	Depth (m)	Bernoulli Hump height (cm)
1	5 (2.57)	10	10.14
2	5 (2.57)	20	1.97
3	5 (2.57)	30	0.97
4	5 (2.57)	40	0.58
5	5 (2.57)	50	0.37
6	5 (2.57)	75	0.08
7	5 (2.57)	100	0.06



a)



b)



c)

Fig 9a-c: BH (cm) for Sindhughosh class submarines at 5 knots speed

4. Discussion:

4.1 Way ahead for detection of submarines:

As the previous sections show, submarine leaves barely perceptible signatures on the ocean surface. When the vessel is travelling at 10 knots speed and about 20 m below water, it produces a small (about 15 cm) but measurable Bernoulli Hump. This is measurable by the current altimeter resolutions. This rise and falloff sea surface height is shown in Fig 10 below. The ocean surface as viewed from above (like from a SAR on an airplane) has been recreated in that figure. We can see a perceptible feature at the surface for 20m depth of operation with a BH of 10-15 cm. Beyond that, the chances of detection are low. Over and above this, if sea is rough, these features wouldn't be visible at all. So our work is for calm waters only. For rougher seas, the SAR image wil have to be filtered out to remove the wind waves for the BH to be visible. This is discussed in

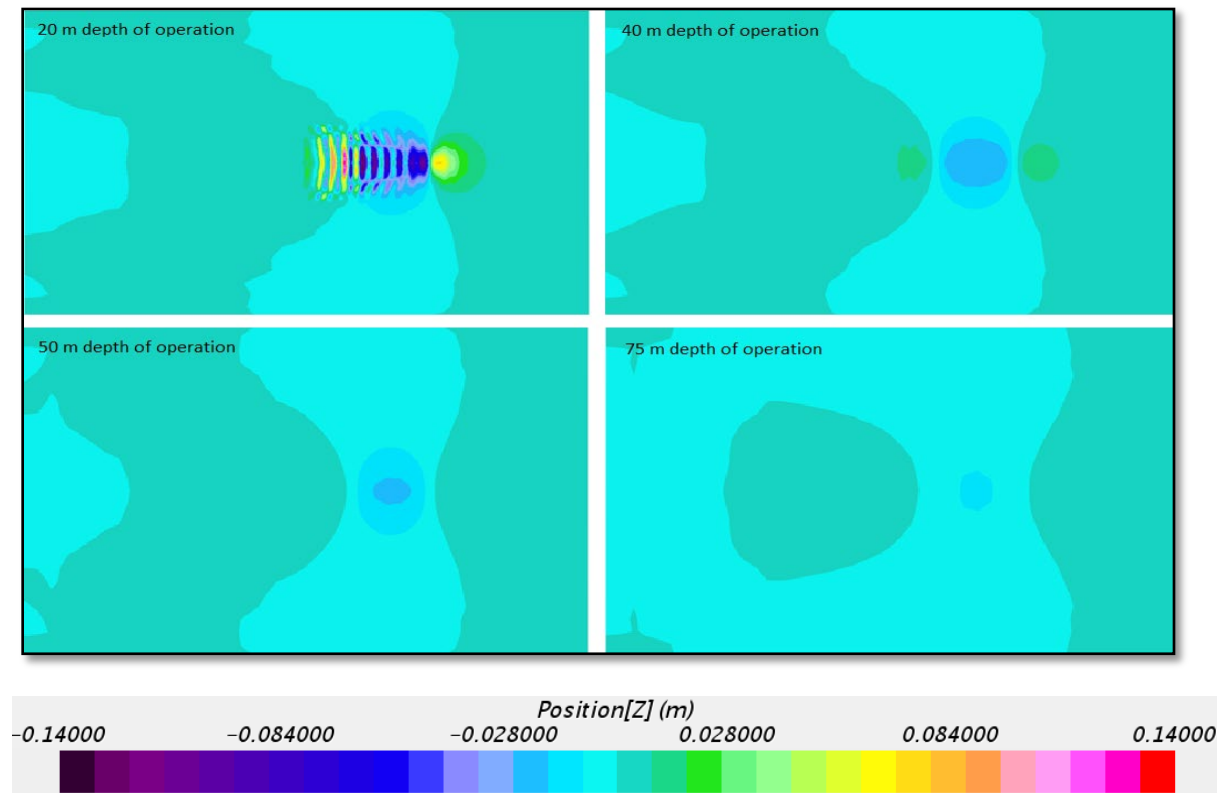


Fig 10: Surface elevation plots for Akula submarine at 10 knots for different depths

short in the next section. This could be compared with an actual SAR image of a submarine surfacing (Fig 11). This is when the submarine is at periscopic depth. As can be seen, there is a very prominent Kelvin wake (a transverse and divergent set of waves making a Kelvin wave angle of 39°). And within it there is a Bernoulli Hump.

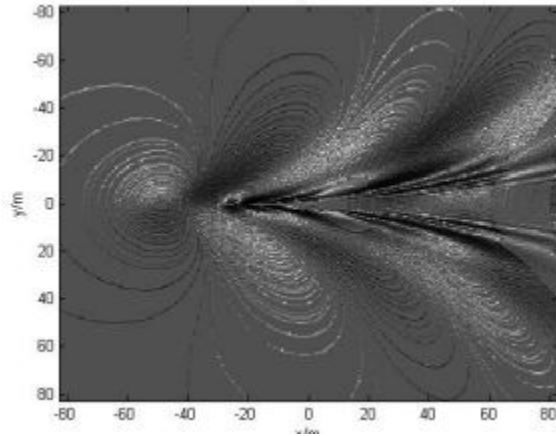


Fig 11: SAR image of submarine at periscopic depth.

4.2 BH in presence of regular waves:

It is a known fact that ocean can never be in a totally calm state without environmental waves. Such waves can be imposed on a submarine wake thus, totally obscuring the BH. To detect and bring to use the properties of BH in such cases, we need to filter out these environmental waves. Ocean exists as a spectrum of waves (like JONSWAP, Pierson-Moskovich or Bretschneider spectrum). Conventionally, the significant wave heights and periods are forecasted from the wind information on the basis of the wind-wave relationship. However, the error may become large due to many uncertainties in the wind generation prediction and wind-wave relationship..

This paper is mainly focused on calm-wave conditions and we have only made a foray into the presence of environmental waves. Quite a bit of work is already completed, but mainly in terms of regular waves. Regular waves can easily be filtered out by using a filter on FFT analysis. Thus the periodic waves are removed by FFT filtering. The purpose of doing filtering is to get the BH in calm water, which can then be used for reverse calculation of submarine particulars (work under progress). This, we include in the paper.

Here, the waves and Bernoulli hump exist merged in the space domain. Wavenumber distribution of the space domain is found by Fast Fourier Transform (FFT). By using the Band-stop filter of appropriate range, filtering is done on the wavenumber domain. Then, Inverse FFT is applied to the filtered wavenumber domain, to get the wave subtracted free surface in which only BH is present. If there is more than one wave, the same procedure of applying Band-stop filters to the wavenumber domain can be followed. Thus, for these regular waves, FFT is employed on the sea surface elevations. To do FFT, the freesurface should be uniform grid spacings, which it is (Fig 2b). Currently we have done it only for regular waves at the surface. Figure below shows the wave cut at the centerline of the submarine in the free-surface with a simulated Bernoulli Hump of submarine moving at 5.144 m/s at the water depth of 10 m with waves of wavelength 80 m and wave amplitude 0.1 m. Filtering was done using the 2D FFT routine of Matlab.

The final goal of these studies is to reverse-calculate the details of the submarine from the features of Bernoulli Hump (its depth variation and length). For this, it is enough if we study a longitudinal cut along the length of the submarine at the free surface above the centerline. We measure the BH parameters along that line which can then be fed into reverse-calculation algorithms. So we show the Fig 12 for that wave-cut only. But this can easily be extended to the whole surface.

Procedure

1. Obtain the free surface elevations of ocean waves along the wave cut where the wave subtraction has to be done (Fig 12).
2. Apply Fast Fourier Transforms on the free surface and plot wave number distribution of waves in it.

3. By using the appropriate Filter, filter out the waves of particular wave numbers.
4. Apply Inverse Fast Fourier Transforms on the filtered wavenumber distribution, to retrieve the final free surface elevation (Fig 13).

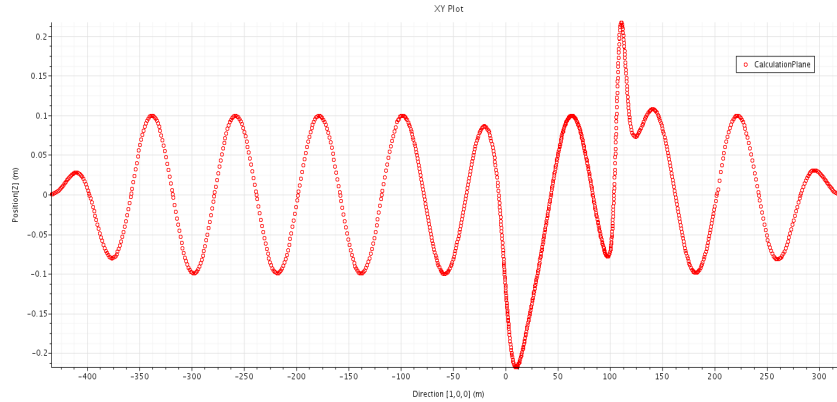


Fig 12: Wave elevation plot above the position of submarine. BH can be seen at the position of 100 m.

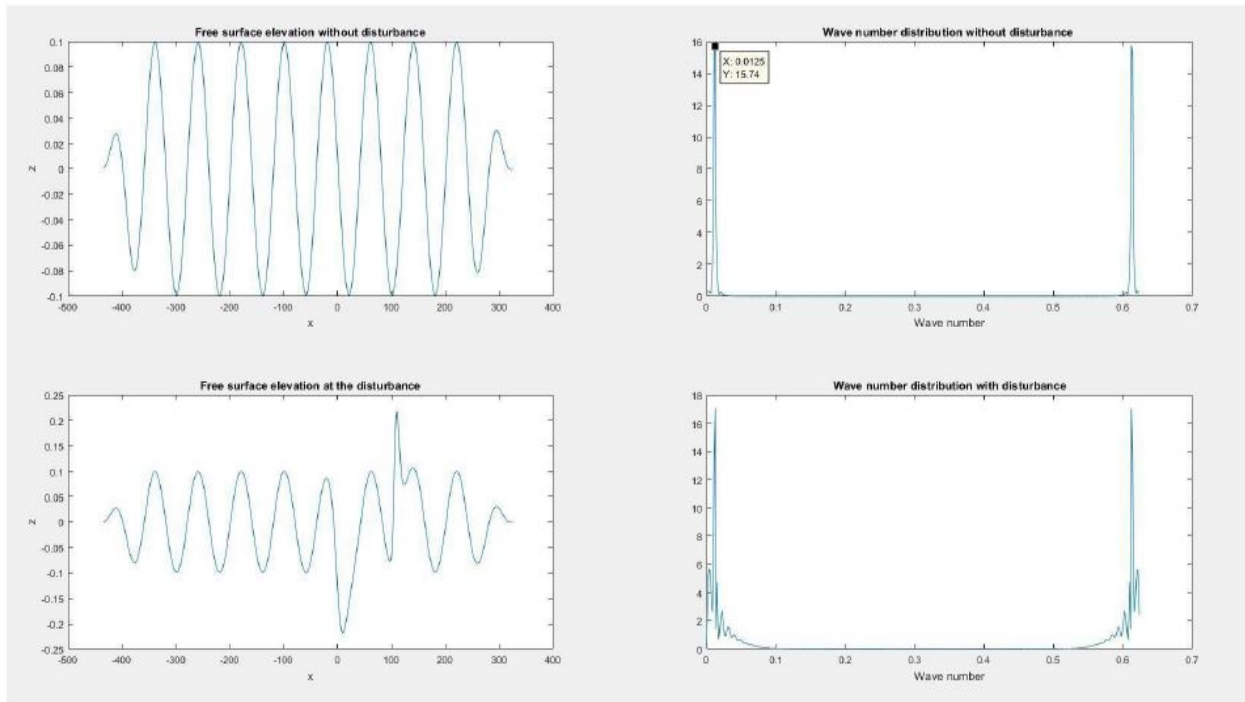


Fig 13: Matlab simulation of FFT filtering.

5. CONCLUSIONS

In this paper, we use an indirect method of detecting submarines by processing the environmental anomalies produced while the vessel is near the surface. Numerical and experimental investigations were performed successfully and it is concluded that:

- Both experimental and numerical analysis converged towards accuracy and shows that the detection of submarines using these techniques is robust and stable.
- In the numerical simulations, two submarines of Indian Navy (Akula class submarine and the Sindhughosh class submarine) have been taken for study. These two submarines have been simulated for near-surface motion at 5 knots and 10 knots respectively. At 10 knots speed, the Akula submarine shows a BH of 162 cm which can easily be detected by SAR. By the time it goes to 20 m depth, the BH height is only about 15-18 cm.
- BH can also be seen in the presence of environmental waves. The wind waves need to be filtered out by FFT filtering using Matlab toolbox.
- A false image of sea surface elevations has been created for Akula class submarine for various depths of operation. This is what we would see from SAR images. Image processing of the SAR images can therefore be a method of detection of the submarine.
- Reverse calculation schemes are currently underway to study the submarine particulars from the above SAR images.

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REFERENCES

- ASME V&V20, 2006, Standard for Verification and Validation in Computational Fluid Dynamics and Heat Transfer, ASME.
- Barman, J., Kumar, A., Warrior, H.V., Maheswaran, P.A. and Kumar, H. Wide area surveillance using ship wakes. *CURRENT SCIENCE*, 114(8), pp.1606-1607, 2018.
- Benedict L, R. Gould, Towards better uncertainty estimates for turbulence statistics Exp. Fluid, 22 (2) (1996), pp. 129-136
- Birkhoff, G and E. H. Zarantonello. *Jets, wakes, and cavities* New York: Academic, 1957.
- Brogia R, Mascio A. D., Amati G. A. parallel unsteady RANS code for the numerical simulations of free surface flows[C]. 2nd international Conference on Marine Research and Transportation. Naples, Italy, 2007.
- Celik I.B., U. Ghia, P.J. Roache, C.J. Freitas, H. Coleman, P.E. Raad, Procedure for estimation and reporting of uncertainty due to discretization in CFD applications, J. Fluid Eng. Trans. ASME, 130 (2008), pp. 078001-078004
- Chen J, B. Chen and J. Yang, "The Turbulent Wake Detection and Beam Estimation of a Ship in SAR Images," 2008 International Workshop on Education Technology and Training & 2008 International Workshop on Geoscience and Remote Sensing, 2008, pp. 230-233, doi: 10.1109/ETTandGRS.2008.157.
- Cimbala, J.M. Experimental uncertainty analysis. *Online Edition*, pp.1-4, 2013.
- Deng Y, H. Wang, S. Liu, M. Sun, X. Li , Analysis of the ship target detection in high-resolution SAR images based on information theory and Harris corner detection, EURASIP J. Wirel. Commun. Netw., 2018 (2018), p. 291
- Daly D.G, "A limited analysis of some non-acoustic antisubmarine warfare systems", M.Sc. Thesis, Naval Postgraduate School, Monterey, California, March 1994.
- Dubbioso G, R Brogia, S Zaghi, CFD analysis of turning abilities of a submarine model, Ocean Engineering (2016) 129, 459-479.
- Feldman J, Straightline and Rotating Arm Captive-model Experiments to Investigate the Stability and Control Characteristics of Submarines and Other Submerged Vehicles David Taylor Research Center, Bethesda MD, Ship Hydromechanics Department (1987)
- M. Gibson, B. Launder, Ground effects on pressure fluctuations in the atmospheric boundary layer, J. Fluid Mech., 86 (3) (1978), pp. 491-511

Griffin, O.M., Wang, H.T. and Meadows, G.A. Ship hull characteristics from surface wake synthetic aperture radar (SAR) imagery. *Ocean engineering*, 23(5), pp.363-383, 1996.

Hershey A V, "Measured versus computed wave trains of a Rankine ovoid", U.S. Naval Weapons Laboratory, Dahlgren, Virginia Technical Report, AD 641888, 1966.

HU Zhi-qiang, Yang LIN, Haitao GU, On Numerical computation of viscous hydrodynamics of unmanned underwater vehicle[J], *Robot*, 29 (2) (2007), pp. 145-150, (in Chinese).

Jagadeesh P, K. Murali, V. Idichandy. Experimental investigation of hydrodynamic force coefficients over auv hull form *Ocean Eng.*, 36 (1) (2009), pp. 113-118

Kanjir U, H. Greidanus, K. Oštir, Vessel detection and classification from spaceborne optical images: a literature survey, *Remote Sens. Environ.*, 207 (2018), pp. 1-26

Kundu, P., Sarkar, A. and Nagarajan, V. Improvement of performance of S1210 hydrofoil with vortex generators and modified trailing edge. *Renewable Energy*, 142, pp.643-657, 2019.

Mansoorzadeh S, E. Javanmard, An investigation of free surface effects on drag and lift coefficients of an autonomous underwater vehicle (auv) using computational and experimental fluid dynamics methods, *J. Fluid Struct.*, 51 (2014), pp. 161-171

Mitra A, J. Panda, H. Warrior, The effects of free stream turbulence on the hydrodynamic characteristics of an auv hull form, *Ocean Eng.*, 174 (2019), pp. 148-158

Munk Walter Heinrich, P. Scully - Power and F. Zachariasen, The Bakerian Lecture, 1986 Ships from space, *Proceedings of the Royal society of London, A, Mathematical and Physical sciences* (1987), vol 412, Issue 1843

Newman, J.N. The inverse ship-wave problem. In *Sixth International workshop on water waves and floating Bodies* (pp. 193-197), April, 1991.

Panda J, A. Mitra, A. Joshi, H. Warrior, Experimental and numerical analysis of grid generated turbulence with and without mean strain *Exp. Therm. Fluid Sci.*, 98 (11) (2018), pp. 594-603

Peltzer R, J. H. Milgram, R. Skop, J. Kaiser, O. Griffin and W. Barger, "Hydrodynamics of ship wake surfactant films" in *Proc. 18th Symp. Naval Hydrodynamics*, Washington, DC: Nat. Acad. Press, pp. 533-552, 1991.

Peregrine, D.H. A ship's waves and its wake. *Journal of Fluid Mechanics*, 49(2), pp.353-360, 1971.

Quick H, R. Widjaja, B. Anderson, B. Woodyatt, A.D. Snowden, S. Lam, Phase II Experimental Testing of a Generic Submarine Model in the DSTO Low Speed Wind Tunnel, DSTO Technical Note, DSTO-TN-1274 (2014)

Rabaud M and Frederic Moisy, Ship Wakes: Kelvin or Mach Angle?, Physical Review Letters, PRL 110, 214503.

Sakthivel R, S. Vengadesan, S. Bhattacharyya, Application of non-linear κ - ϵ turbulence model in flow simulation over underwater axisymmetric hull at higher angle of attack, J. Nav. Architect. Mar. Eng., 8 (2) (2011), pp. 149-163

Saeidinezhad A, A. Dehghan, M.D. Manshadi, Experimental investigation of hydrodynamic characteristics of a submersible vehicle model with a non-axisymmetric nose in pitch maneuver, Ocean Eng., 100 (2015), pp. 26-34

Stefanik, T., The Nonacoustic Detection of Submarines, Scientific American, March 1988, p.29.

Scully-Power, P. and Stevenson, R.E., Swallowing the Transparency Pill, Proceedings/ December 1987, p.151

Thomson, W. On ship waves. *Proceedings of the Institution of Mechanical Engineers*, 38(1), pp.409-434, 1887.

Tunaley, The Bernoulli Hump Generated by a Submarine, March 2015, DOI:10.13140/RG.2.1.3278.0322, Report number: LRDC 2015-03-001, Affiliation: London Research and Development Corporation.

Tunaley, J.K. The estimation of ship velocity from SAR imagery. In *IGARSS 2003. 2003 IEEE International Geoscience and Remote Sensing Symposium. Proceedings (IEEE Cat. No. 03CH37477)* (Vol. 1, pp. 191-193). IEEE, July, 2003.

Tunaley, J.K., Buller, E.H., Wu, K.H. and Rey, M.T. The simulation of the SAR image of a ship wake. *IEEE transactions on geoscience and remote sensing*, 29(1), pp.149-156, 1991.

Tyagi A, D. Sen, Calculation of transverse hydro-dynamic coefficients using computational fluid dynamic approach[J] *Ocean Engineering*, 33 (5–6) (2006), pp. 798-809

Voulgaris G, J.H. Trowbridge, Evaluation of the acoustic Doppler velocimeter (adv) for turbulence measurements, *J. Atmos. Ocean. Technol.*, 15 (1) (1998), pp. 272-289

WILCOX, D. C. (1993). *Turbulence modeling for CFD*. La C  nada, CA, DCW Industries, Inc

WU Ban-shan , Fu XING, Xiao-feng KUANG, *et al.* Investigation of hydrodynamic characteristics of submarine moving close to the sea bottom with CFD methods[J], *Journal of Ship Mechanics*, 9 (3) (2005), pp. 19-28

YAN Yong, G, Zao-jian ZOU, Chen-xi ZHANG, Calculation of hydrodynamic forces on a KVLCC hull in sway motion in deep and shallow water[J], *Chinese Journal of Hydrodynamics*, 26 (1) (2011), pp. 85-93

Zhang, Z.R. Verification and validation for RANS simulation of KCS container ship without/with propeller. *Journal of Hydrodynamics*, 22(1), pp.889-896, 2010.

Zilman, G., Zapolski, A. and Marom, M. The speed and beam of a ship from its wake's SAR images. *IEEE Transactions on Geoscience and Remote Sensing*, 42(10), pp.2335-2343, 2004.

Zilman, G. and Shemer, L. An exact analytic representation of a regular or interferometric SAR image of ocean swell. *IEEE transactions on geoscience and remote sensing*, 37(2), pp.1015-1022, 1999.